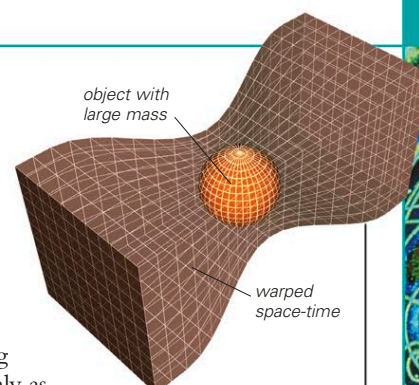


DENTED SPACE-TIME

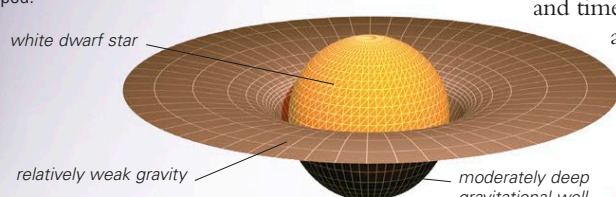
Space-time can be visualized as a rubber sheet in which massive objects make dents. In this view, planets orbit the Sun because they roll around the dent it produces. Similarly, light passing by a massive object has its straight-line path deflected by following the local curvature of space-time. Remember, however, that it is four-dimensional space-time, not just space, that is warped.

GENERAL RELATIVITY AT WORK

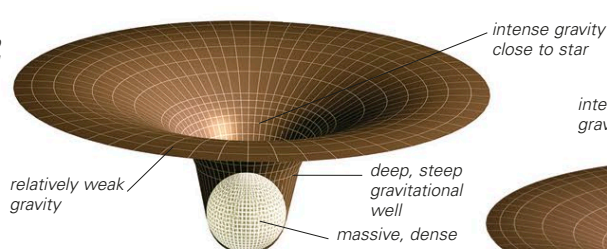
Einstein encapsulated his theory of how mass distorts space-time in his set of “field equations.” Physicists have used these equations to find that it is in the strongest gravitational fields—where massive, dense objects distort space-time most strongly—that reality departs furthest from that predicted by Newton (see p.38). For instance, Mercury is so close to the Sun that it always moves in a strong gravitational field (or in strongly curved space-time). Its orbit is distorted in a way that Newton could not account for, but which general relativity explains perfectly (see p.110). General relativity also provides a framework for models of the universe’s structure, development, and eventual fate. It predicts that the universe must be either expanding or contracting. Before the introduction of general relativity, space and time were thought of only as an arena in which events took place. After general relativity, physicists realized that space and time are dynamic entities that can be affected by mass, forces, and energy. However, while general relativity accurately describes the universe on a large scale, it has little to say about how it works at the tiniest, subatomic scale.

**PINCHED SPACE**

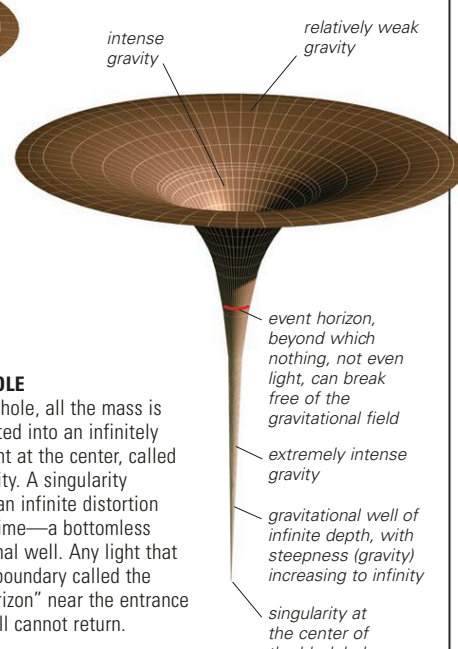
Instead of a two-dimensional sheet, four-dimensional space-time can also be visualized as a three-dimensional volume that is narrowed or “pinched in” around large masses.

**WHITE DWARF STAR**

A white dwarf is a very dense, planet-sized star that can be thought of as producing a smaller but deeper dent in space-time than a star like the Sun does.

**NEUTRON STAR**

A neutron star is an exceedingly dense stellar remnant that makes a very deep dent in space-time. A neutron star significantly deflects light passing by, but cannot capture it.

**BLACK HOLE**

In a black hole, all the mass is concentrated into an infinitely dense point at the center, called a singularity. A singularity produces an infinite distortion in space-time—a bottomless gravitational well. Any light that passes a boundary called the “event horizon” near the entrance to this well cannot return.

GRAVITATIONAL WAVES

One of Einstein’s major predictions based on general relativity was the existence of gravitational waves. These waves can be regarded as disturbances in the curvature of space-time that propagate outward from their source—accelerated masses—at the speed of light. For over 100 years, gravitational waves remained hypothetical and unobserved, but in early 2016, it was announced that the first direct observation of such waves had been made, originating from a pair of merging black holes some 1.3 billion light-years from Earth. The waves were detected by a collaboration between scientists working on an experiment at a US observatory called LIGO (Laser Interferometer Gravitational-Wave Observatory) and others operating the Virgo interferometer, an instrument in northern Italy that is also designed to identify gravitational waves. Since 2016, several further instances of gravitational waves have been observed emanating from merging black holes or, in one case, from merging neutron stars (see p.267).

MERGING STARS

In this artist’s rendition, two neutron stars orbit each other at high speed as a prelude to merging. Such an event is predicted to produce both gravitational waves and a burst of gamma rays—both detected in August 2017 by the LIGO-Virgo project.

GRAVITY BENDING LIGHT

The effect of gravity on the path of light is not obvious unless an observer looks deep into space at the universe’s most massive objects—clusters of galaxies. This image shows galaxies as white blobs. Their combined gravity bends light so much that the images of more distant galaxies appear as blue streaks, stretched and squashed by the galaxy cluster’s gravity.

